

# expression\_template [↗](#)

Module to handle the parsing and conversion of a Pulse ExpressionTemplate containing Pulse schema fields and a string expression.

## ExpressionTemplateJava module-attribute [↗](#)

```
ExpressionTemplateJava = autoclass(MODEL_CLASSPATH + 'datascience.ExpressionTemplate')
```

## ExpressionTemplate [↗](#)

```
ExpressionTemplate(fields: List, expression: str, raw_expression: str)
```

Util class to handle conversion from expression templates to indexed strings and vice versa.

## FIELD\_PATTERN class-attribute instance-attribute [↗](#)

```
FIELD_PATTERN = '\\$\\{(.+?)\\}'
```

## INDEX\_PATTERN class-attribute instance-attribute [↗](#)

```
INDEX_PATTERN = '\\$\\{((\\d+)\\}'
```

## expression instance-attribute [↗](#)

```
expression = expression
```

## fields instance-attribute [↗](#)

```
fields = fields
```

## raw\_expression instance-attribute [↗](#)

```
raw_expression = raw_expression
```

## from\_java\_template classmethod [↗](#)

```
from_java_template(exp_template: JavaClass) -> ExpressionTemplate
```

Converts a Pulse API ExpressionTemplate to a string with indexed fields.

If a field is used multiple times the field is duplicated and the index incremented.

Parameters:

| Name                                        | Type                            | Description                  | Default         |
|---------------------------------------------|---------------------------------|------------------------------|-----------------|
| <code>exp_template</code> <a href="#">↗</a> | <code>ExpressionTemplate</code> | Pulse API ExpressionTemplate | <i>required</i> |

Returns:

| Type                            | Description        |
|---------------------------------|--------------------|
| <code>ExpressionTemplate</code> | ExpressionTemplate |

### `from_string` classmethod [¶](#)

```
from_string(indexed_str: str)
```

Creates a new ExpressionTemplate from a string with indexed fields.

Fields must be enclosed with the pattern `${field}`.

Parameters:

| Name                                       | Type             | Description                                                                 | Default         |
|--------------------------------------------|------------------|-----------------------------------------------------------------------------|-----------------|
| <code>indexed_str</code> <a href="#">¶</a> | <code>str</code> | String with indexed fields eg. <code>"\${TransactionAmount} &gt; 10"</code> | <i>required</i> |

Returns:

| Type                            | Description        |
|---------------------------------|--------------------|
| <code>ExpressionTemplate</code> | ExpressionTemplate |

### `rename_fields` [¶](#)

```
rename_fields(old_name: str, new_name: str)
```

Renames all the fields matching the `old_name`.

Parameters:

| Name                                    | Type             | Description              | Default         |
|-----------------------------------------|------------------|--------------------------|-----------------|
| <code>old_name</code> <a href="#">¶</a> | <code>str</code> | Old name to change from. | <i>required</i> |
| <code>new_name</code> <a href="#">¶</a> | <code>str</code> | New name to change to.   | <i>required</i> |

### `to_java` [¶](#)

```
to_java() -> JavaClass
```

Converts to DS-API object.

Returns:

| Type                   | Description                   |
|------------------------|-------------------------------|
| <code>JavaClass</code> | ExpressionTemplate Java class |